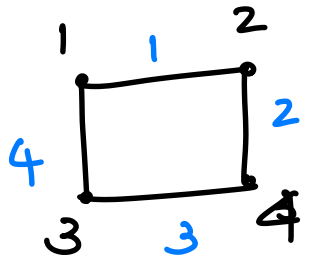
Spanning Tree

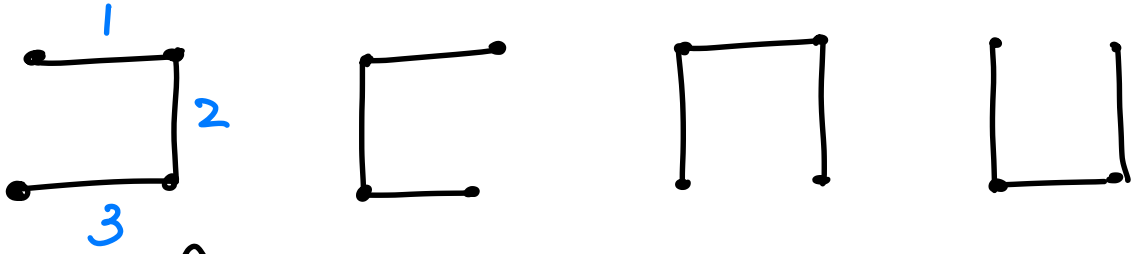
Tree → Connected
Tree → Acyclic

Spanning Tree → A subgraph which covers all the vertices.

eg:



Spanning Trees:



wt = 6

Min wt spanning tree

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Brute Force:

Find all spanning trees

Calculate weights

Choose min

Q: Given a connected graph G with n vertices, how many spanning trees of G are possible?

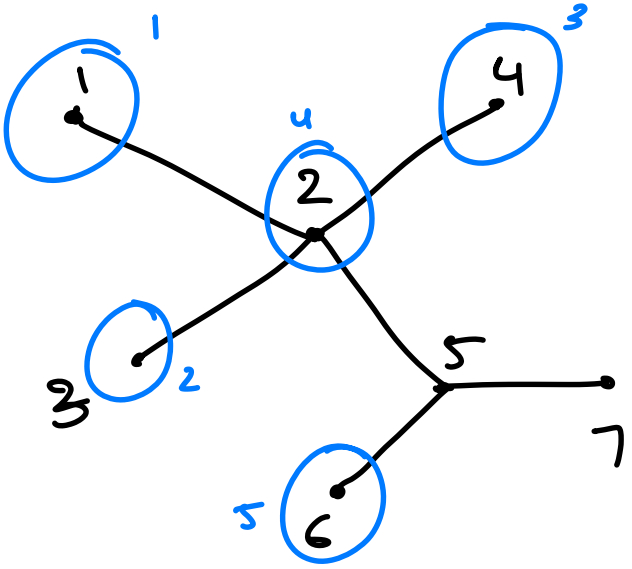
Prufer Sequence

Remove the leaf with Smallest label.

The 1st term of the sequence is the label of leaf's neighbour

Repeat this process

(1)



5 7

leaves

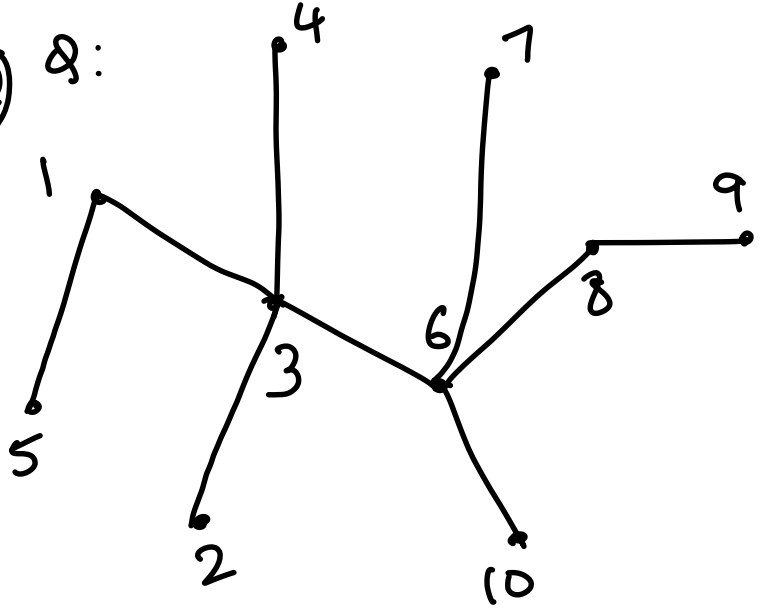
$\{2, 2, 2, 5, 5\}$ is prüfersequence.

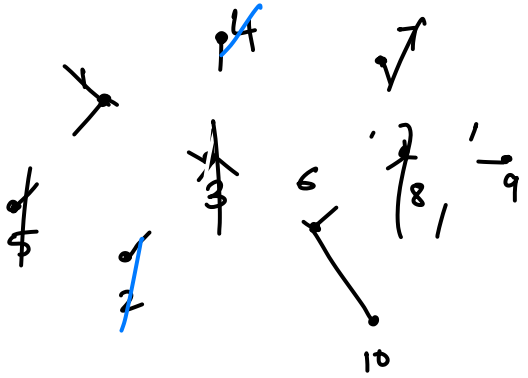
1's neigh

3's neighbour

4 2 6

(2) 8:





$\{3, 3, 1, 3, 6, 8, 8, 6\}$

$\{2, 2, 2, 5, 5\}$ ($n-2$)

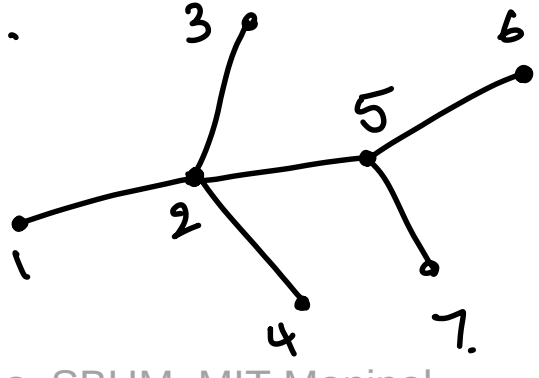
$\hookrightarrow n-2=5$

∴ 2 vertices not considered

I) No of vertices $= n-2+2 = 5+2 = 7$.

$S = \{1, 2, 3, 4, 5, 6, 7\}$.

$P = \{2, 2, 2, 5, 5\}$

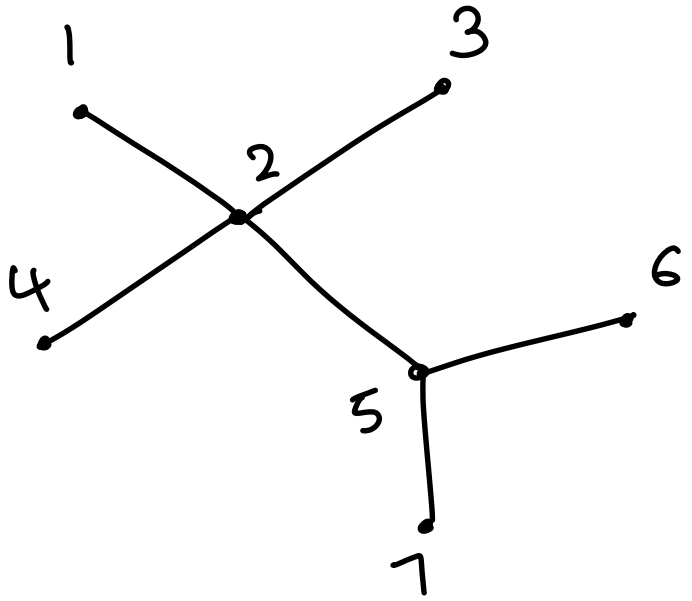


1) Locate 1st number in S that is not in P and connect it with first elmt of P

Repeat

2) Connect the last remaining (2) vertices in S

.



Labelled

Seq^n $\longleftrightarrow$ Tree

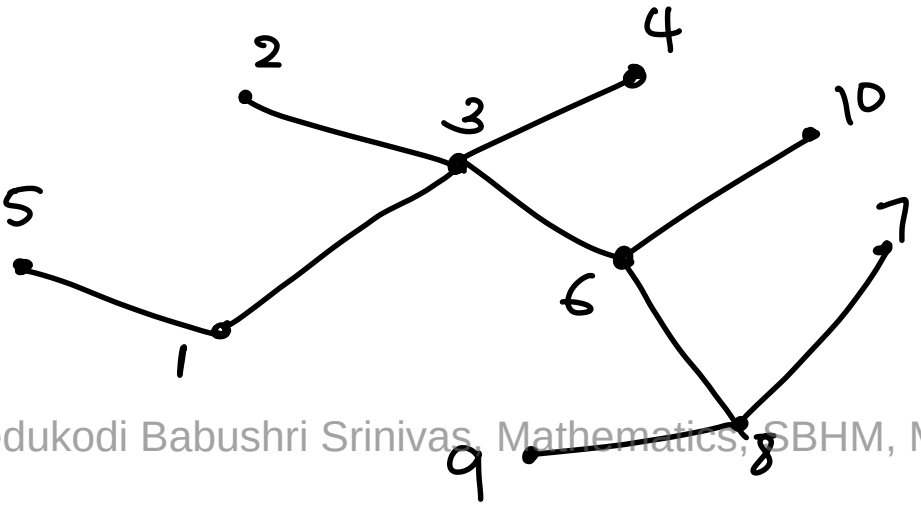
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$P = \{3, 3, 1, 3, 6, 8, 8, 6\}$

$n - 2 = 8$

$n = 10$

$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$



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Result:

There are n^{n-2} labelled spanning trees given a graph with n vertices

Proof:

Take a spanning tree T of the graph G.

$\Rightarrow$ T has n vertices.

$\Rightarrow$ Prufer sequence of T contains $n-2$ elements. say

$$P = \{ b_1, b_2, \cdots, b_{n-2} \}$$

Each element b_i can be $1 \leq b_i \leq n$. (from 1 to n)

$\Rightarrow$ Each b_i has n-choices.

Total number of choices

$1 \le i \le n$.

$\Rightarrow$ for all $b_i = \underbrace{n \cdot n \cdots n}_{n-2 \text{ times}}$

$= n^{n-2}$

$\therefore$ By brute force, finding spanning trees is exponential

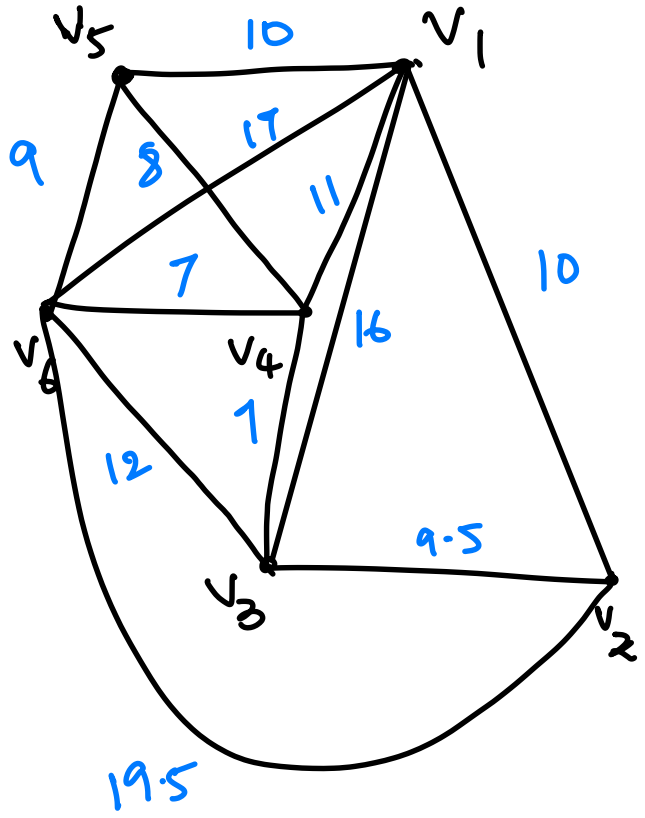
Prim's Algorithm: (Minimum Spanning tree)

1. Build a tree one vertex at a time

2. Start with a trivial one point tree T

   Then add the edge of minimum weight among those with one point in T and the other not in T.

Eg:



$V_1 = \{10, 17, 11, 16, 10\}$

$V_5 = \{10, 8, 9\}$

$V_4 = \{11, 7, 8, 7\}$ ∴ among all 7 is least.

$V_6 = \{12, 19.5\}$

$V_3 = \{12, 11, 16, 9.5\}$

$V_2 = \{10, 17, 11, 16, 10\}$

$V_1V_5 - 10$

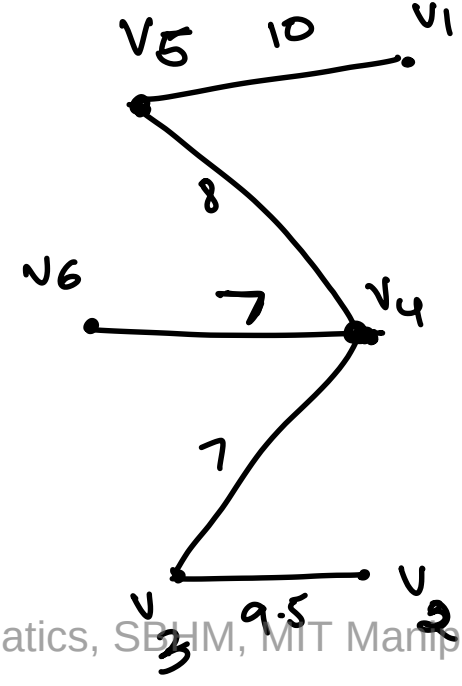
$V_5V_4 - 8$

$V_4V_6 - 7$

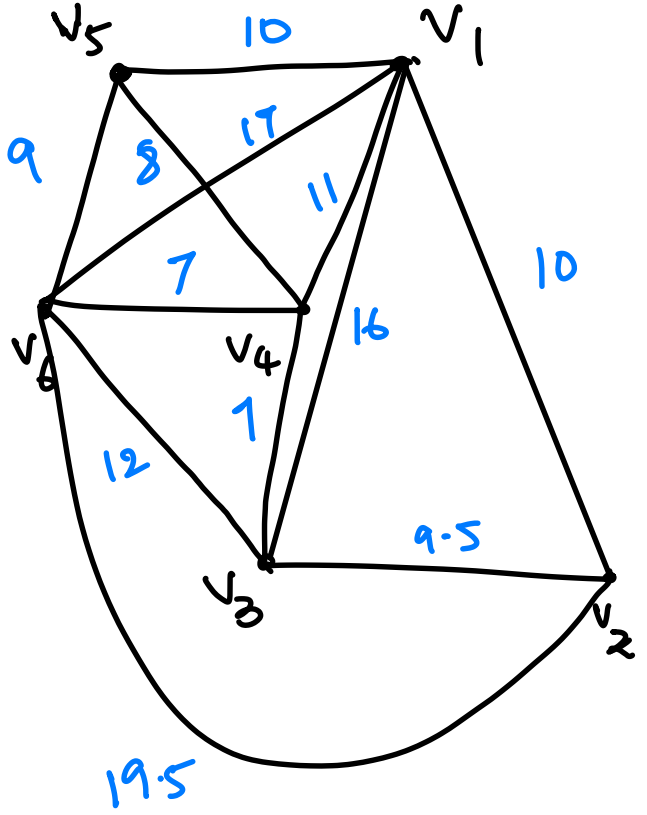
$V_4V_3 - 7$

$V_3V_2 = 9.5$

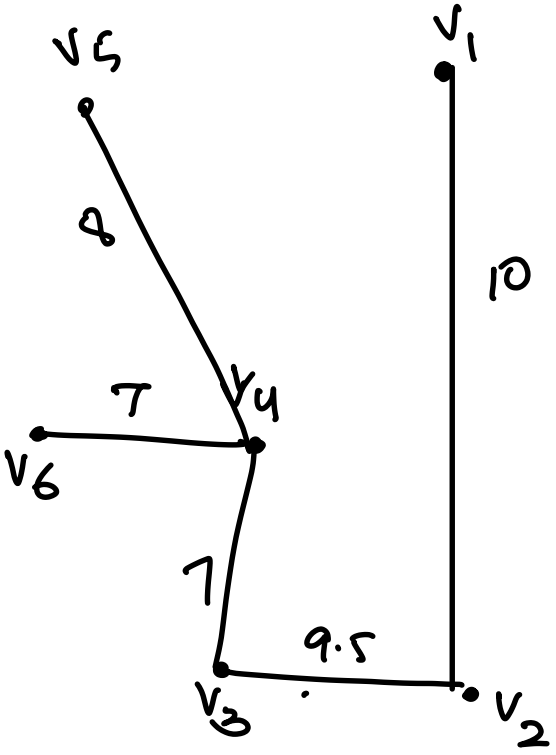
Minimum Spanning Tree:



weight $= 9.5 + 7 + 7 + 8 + 10 = 41.5$



Another)



$10+8+7+7+9.5=41.5$